

Module 7: Risk & Return

Topic 127: Risk & Return Analysis

Return

- **Additional amount on investment**
- **Dividend**
- **Interest**

Risk

- **Uncertainty**
- **Expected Return/Mean**
- **Deviation for Mean/Std. Dev.**
- **Risk & Return are Proportional**

Risk and Return are two concepts which go together and in hand. It means that Risk and Return are two such concepts which are difficult to understand separately but we need to understand them with each other's reference.

Return:

When we define **Return**, it is simply the **additional amount** that **we get on our investment**. When we use the term Return, the other term that is very obvious and that should be discussed along with Return is Investment.

Investment means that **if we forgo or give our money to someone or invest anywhere for some time** against which **we get some additional amount or additional return after some time**. And this is what we say is the concept of Return.

So, **when we say that we are earning some Return**, it means **we have made some investment and the rate of return or the amount of return is certainly that every investor would consider**.

The investor receives this return in which form and for this we need to know the two basic kinds of investment.

One investment may be in equity meaning that we purchase shares of any business organization or company and become owner of that company as we know that the shareholders are the owners of a company. And the other possibility is that we give a loan or debt to the company and we become as debt holder. A shareholder is an investor and a debt holder is also an investor.

How can an Individual be a lender to a corporation or a business organization?

Individuals purchase bonds of the companies or debt certificates or debt securities of a company to become lenders and investors in the form of debt to the company.

The people or investors who purchase shares or stocks of any company then the return that they get are profit. And the profit when distributed among these shareholders then it is called Dividend. So, the Dividend is such return that is given to the shareholders on their investment.

Likewise, the lenders, the bond holders, the debt security holders; the return that they get is generically called Interest. So, Return is that additional amount which will be given to the investor and will be in the form of dividend or interest.

Risk:

Risk actually is the name of Uncertainty. Risk means that when we are investing our money how unsure we are that what will be the return that we will get. If we are sure that the rate of return or the amount of return that we will get on our investment will be exactly the same as we have anticipated then there is no Risk.

But this is not the case whenever we are making investments. Particularly, when we are talking about the Stock market, Security market, Capital market and other types of markets where most of the investors put their money. So, Risk is Uncertainty of Return.

When we talk about Risk with reference to the investment in shares and in bonds, actually initially there is an Expectation of Return from our Investment. When any

investor invests his money anywhere then he has some expectation of return that he wants to get from this investment. He sees the past returns of that business or company and by calculating percentage of an average return to develop or make it his expectation. Meaning that **Expected Return on an investment** is actually the average or mean return on that investment of the past few years or month or weeks or days even.

And when we talk about Risk, it is the Deviation from that Expected Return. It means that Risk is that our Expectation which we have to get return then our actual return can be how much more or less than that Expectation. Because in reality, we cannot be sure that our Expected Return or our Actual return both will be the same.

So, the Deviation or possible Deviation or probable Deviation from our Expected Return which is the Mean, is actually the Risk.

Now, the last thing that we need to understand and remember about these two concepts of Risk and Return that these are **Directly Proportional** to each other.

What is meant by Directly Proportional?

It means that if one thing will increase then other will also increase and similarly if one thing will decrease then other thing will also decrease. It means that if we want that our Return increase then we have to take more Risk. And if we decrease our Risk level then our Return will also be decreased.

So, the securities where the Risk Level is high, provide higher Return and the securities where the Risk Level is low also provide low of Return.

To conclude our discussion on these concepts of Return & Risk analysis, we would say that **Return is what we get on our investment.** But the question that we need to remember that this Return can be positive as well as negative and that is something which is related to Risk and will discuss in detail in upcoming lectures.

Module 7: Risk & Return

Topic 128: Systematic and Unsystematic Risk

Risk

Market Risk

- **Undiversifiable Risk**
- **Unavoidable Risk**
- **Systematic Risk**
- **Volatility**

Unsystematic Risk

- **Avoidable Risk**
- **Diversifiable Risk**
- **Controllable Risk**

Market Risk:

In Stock Market or Security Market, basically there are different types of Equity and Debt Securities which are traded. And on these securities, share or bonds or certificates there is an element of Risk.

When talk about Stock Market, in the Stock Market because of various securities that is shares and bonds then there is a Risk in the Market. It means that an internal Risk become created due to the presence of all these shares in the market that we call Market Risk. It means that the investor, who will invest money in Stock Market, will have to expose from that market risk and we also name it Unique Risk.

In Stock Market or in Security Market, there is built-in Risk. The investor who will invest money in that stock market then he must have to expose from that Risk and he cannot be saved from that Risk. And only can be saved if he has not to invest in that market. So, the Risk which is the Risk of the market is the Systematic Risk and we name it Market Risk also. And we also name it Unique Risk or Undiversifiable Risk.

Here the term **Diversification** means not putting all your eggs in one basket. This is a phrase which is very commonly used when describing Diversification. It means that investor has not invest all amount of his investment in one security or share or bond of a company but divides it into different shares or bonds or securities and this is called **Diversification**.

As the Stock Market is more diversified in which hundreds of company's shares or bonds are traded therefore investor need to do diversification in order to control his risk in that well diversified market. But maximum diversification an investor can do is to put the money in all the companies which are trading in the stock market. Then its diversification becomes equal to the diversification of the stock market but cannot be more than it. So, the Risk of that market due to the diversification cannot further be reduced and that is why we call it **Undiversifiable Risk**.

That is way is called **Unavoidable Risk**. It means that when we go in stock market then this risk cannot be avoided and that is why the term Unavoidable Risk is used.

And this is what is called in technical term, the **Systematic Risk**. It means that Systematic is risk is that risk which is present in any investment market or stock market and any investor who wants to do trade in that market to earn extra profit cannot do investment without taking that risk.

In stock market, why there is risk?

A term is used for this which we call **Volatility**. The word **Volatile** means no staying on one place, sometimes be more or sometimes less. So, Volatility actually produces risk. This Volatility is price Volatility meaning that any stock's price in the stock market sometimes increases or sometimes decreases and even with a different of fraction of time it increases or decreases. So, this fluctuation is actually called Volatility and the element of risk arises due to this Volatility of stock price in the market. So, the overall Volatility represents the overall risk of market that we call market risk or unique risk or Undiversifiable or systematic risk.

Unsystematic Risk:

Against Systematic Risk, we have such risk that can be **Avoidable Risk** which is other than market risk and that we name **Unsystematic Risk**. It means that if we

say that market risk is Unavoidable then Unsystematic Risk is Avoidable meaning that it is such risk which can be decreased or increased.

Similarly, when we say that market risk or systematic risk is Undiversifiable risk then the risk which we cannot reduce through diversification or investing money in more stocks, it is systematic or market risk. But the risk which we can reduce is **Unsystematic Risk** and that is why we call it **Diversifiable Risk**. It means that if any investor wants for reduce Unsystematic risk then he has to start diversification.

Unsystematic risk is over and above than Systematic risk means that in addition to the Systematic risk there is Unsystematic risk and that is why it can be diversified meaning that if we say that what is my total risk in investment then that will be systematic risk plus unsystematic risk.

So, **Unsystematic risk is Avoidable Risk and it is Diversifiable Risk.**

We can conclude our discussion related to Systematic and Unsystematic risk that when investor invests his money in the stock market then he has to face risk and to face that risk, he has two types of risks; one is that he has to take in any condition which is market risk and other is over and above risk which he can control or limit the we name Unsystematic risk. And then we said that Diversification is the tool that can help us to reduce our Unsystematic risk.

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Topic 129: Individual Stock Risk Analysis-1

Individual Stock's Risk

In this lecture, we will talk about Risk of an Individual Stock and this is the first part of our discussion.

Individual Stock's Risk is actually the Standard Deviation of the Individual Stock's return. We can calculate the risk and expect return of Individual Stock or Portfolio or whole Stock Market. So, if we want to calculate the whole market's expected return then that will be the **Average or Mean return of that stock market.**

Similarly, if we are talking about a **Portfolio** (a Portfolio is a group or combination of various types of securities that is shares and bonds which are combined as an investment and then investor tries to earn return from that Portfolio) then its average return, expected return will be its mean return.

Here we are focusing on the **Risk & Return of an Individual Stock**. So, we have said that the Expected return on a stock is the Mean which is the average.

What about the risk of an Individual Stock?

The Risk of an Individual Stock is the possible Deviation from this Mean meaning that actually the return can be how much more or less than the value of that Mean. That is the Variation, that is the possible Deviation. So, when we calculate Risk of stock then we use a statistical tool Variance for that and find Standard Deviation through Variance.

Standard Deviation is nothing but the square root of Variance. These are statistical tools and these tools are used in many subjects also. In Investment and Finance, we also make use of those two basic tools of statistics in order to calculate the Return and Risk.

There are generalized formulas of these Mean and Standard Deviation. When we apply these generalized formulas on stock's return, we have to accordingly tailor those formulas.

- **Mean (Expected Return)**

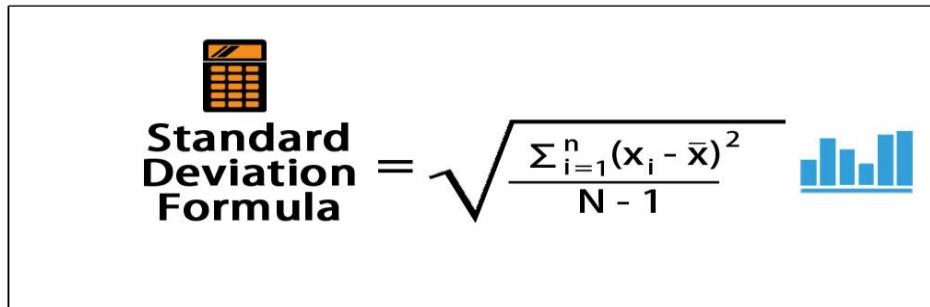
$$\bar{X} = \frac{\sum X}{N}$$

Variance

Standard Deviation

Standard Deviation is possible change or possible deviation or probable deviation from that expected return meaning that in reality the Mean or Average we have calculated then our actual return can be how much more or less than that value of

Average or Mean. Because we have talked that Average return is calculated by adding many returns taken place in the past and dividing by their total numbers.

A graphic showing the standard deviation formula. On the left is a calculator icon. To its right is the text 'Standard Deviation Formula'. This is followed by an equals sign and a square root symbol. Inside the square root is a fraction: the numerator is the sum from i=1 to n of (x_i - x̄) squared, and the denominator is N - 1. To the right of the formula is a bar chart icon with four bars of increasing height.
$$\text{Standard Deviation Formula} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{N - 1}}$$

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Topic 130: Individual Stock Risk-2

In this lecture, we will be talking about Calculations of Stock's Return and Risk and we will see that how this statistical formula that we have talked about are actually applied.

Here we are talking about the Calculation of Mean and Standard Deviation and it is necessary to remember that it means that we are talking about Expected Return and Risk of the Stock.

Average Return: Mean		
Return-R	Probability-P	Average: R x P
-0.1	0.05	-0.005
-0.02	0.1	-0.002
0.04	0.2	0.008
0.09	0.3	0.027
0.14	0.2	0.028
0.2	0.1	0.02
0.28	0.05	<u>0.014</u>
	Average	0.09

Probability of occurrence of return is calculated by considering the frequency of occurrence of a particular return and dividing it by the total number of observations.

Risk: Standard Deviation		
Return-R	Probability-P	Var. = $(R - \text{Ave } R)^2 \times P$
-0.1	0.05	$\{(-0.1 - 0.09)^2\} \times 0.05$ = 0.001805
-0.02	0.1	0.00121
0.04	0.2	0.0005
0.09	0.3	0
0.14	0.2	0.0005
0.2	0.1	0.00121
0.28	0.05	<u>0.001805</u>
	Variance	0.00703
	Std. Dev.	0.0838 or 8.38%

When we conclude our discussion then we can say that we are expecting on this investment whose data we have used in the column of Return that our Return will be 9% and from this expectation our actual result can deviate 8.38% meaning that

we can say that if we add 8.38% in 9% then 17.38% can be our maximum return or if we subtract 9% from 8.38% then 0.62 can be our minimum return and our actual return will fluctuate between that range. So, this is how we calculate the Expected Return and the Risk of an Individual Stock.

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Topic 131: Coefficient of Variation

Coefficient of Variation is a concept that is used to compare and contrast the Risk and Return of more than one Stock. It means that if we want to see that whose risk and return is more and whose risk and return is less by comparing two or four different investments or different stocks or securities then Coefficient of Variation helps us in this way.

There are two types of comparisons which are possible: **Absolute and Relative**.

Absolute means when we say that 10% is less than 20% or 15% is more than 11%, this is actually and Absolute amount or percentage that we are comparing. But then we have to see that if the risk of one stock is there on a particular average return that is being provided then in relation to that expected return is the risk more or less.

Meaning that Individual Risk of any stock in terms of number may be showing more but **is it also more than the reference of its expected return?** This is something which is called Relative Analysis and here Coefficient of Variation helps us a lot.

Coefficient of Variation:

- Coefficient of Variation is a measure of relative dispersion (risk) i.e. per unit of expected return

Formula




$$\text{Coefficient of Variation Formula} = \frac{\text{Standard Deviation}}{\text{Mean}}$$


How would Coefficient of Variation Formula help us?

It help us as an investor to make a decision regarding putting our money in a particular stock. It means that this formula will help us to prioritize that which stock is better than the other stock in terms of its risk and return.

		Company A	Company B
Mean (also known as expected return of a stock)	Expected Return	10%	20%
Std. Dev.	Risk	8%	12%
CV	Relative Risk	80%	60%

So, we can conclude that an investor should not only consider the individual percentage of risk but the more important element and particularly when it comes to the calculation of comparison in making investment in one of the two is the relative risk and which is the Coefficient of Variation.

And in this case, we can see that Coefficient of Variation of Company A is more than Company B then we can conclude that risk level of Company A is more as compare to the Company B. And if anyone wants any suggestion that we have to invest in which company with reference to the level of risk then we would suggest that company A is better than company B.

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Topic 132: Portfolio risk analysis-1

Internal Correlation and Covariance

Correlation:

Correlation is about relationship of two variables means that it is the mutual relationship of two variables.

How we can describe this relationship?

We can say that it can either be a positive relationship or it can be a negative relationship. We will see that what do we mean by having a positive relationship or a negative relationship. But this is how we define the Correlation among the two variables.

Here in our discussion on investments and the returns on Portfolio, we are referring to these two variables as two different stocks. So, we will be talking about the relationship of two different stocks with each other and that is called the Correlation.

As we have said that Correlation can be positive or it can be negative and there may not exist any correlation between the two stocks. How do we calculate this correlation and then what we do afterwards we will discuss it later on in next lectures.

How do we calculate Correlation?

We calculate it by determining the value of a coefficient which is called the correlation coefficient and the symbol that we use for it is ρ (Rho) and Rho describes the value, the level of relationship between two stocks.

Correlation Coefficient - Rho

ρ

It means that as we said that it can be positive or can be negative so, the value of Rho can be from absolute positive to absolute negative meaning we may have got absolute positive 100% correlation or we may have got absolute negative that is -100% correlation. And then it means that we can express our values in the way that our values can be between +100 to -100% or in other words we may say that from +1 to -1 can be the extreme values of correlation coefficient.

Meaning that the maximum or minimum correlation can be +1 or -1 among any two stocks and any value can exist between these numbers. If the value of correlation coefficient of **Rho** becomes **zero** then it means that there does not exist any correlation between any two stocks.

Covariance:

If the correlation will be positive or negative then there will be some effect. Meaning that if the correlation is positive then it is likely that the two variables would move in the same direction or positive correlation means that when there will be variation/deviation/change of stocks then that deviation will be in the same direction.

So, Covariance will depend on the Correlation. It means that **what is the variation of any stock with reference to the variation of other stock** which is **called Covariation or Covariance**.

So, **Correlation is the relationship** and **Covariance is the actual change which takes place because of that relationship**.

It means that if there is positive correlation then we will say that change or covariance among them will also be positive. If we talk in terms of Stocks then one stock is moving positive meaning its price is increasing then other stock's price will also increases. And if the price of one stock is decreasing then other stock's price will also decreases if there exist positive correlation among them.

If we say that this positive correlation is absolute positive then it means that all the changes occurring in the price of stock A then same change in the same direction will also occur in the price of stock B.

And if this correlation becomes negative then their variation will become opposite. It means that if the price of stock A goes up then the price of stock B will go down.

Covariance Measures The Directional Relationship Between The Returns On Two Assets.



When we compare both Correlation and Covariance then we conclude that Correlation is their mutual relationship or it is their internal relationship and that relationship actually reflects their movement, their change. If we purchase such stocks among which there is positive correlation (any two stocks' prices go up or down simultaneously) then it may be that profits may be more or losses may also be more.

But if we include such two stocks among which there is negative correlation then if there will be negative movement or seen loss in one stock then the positive movement of other stock will reduce that loss.

So, in this way, that concept of Correlation and Covariance affects the return of stocks' Portfolio.

We can conclude our discussion that there occurs variation among two stocks on the basis of Correlation and if it is not absolute positive or its value is less than +1 then we will get more return from one stock and will get less return from other stock.

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Topic 133: Portfolio Risk Analysis-2

The important things that we need to keep in mind that there will be more than one stocks in Portfolio (i.e., two, three, five etc.). But the average risk of Portfolio is not equal to the average risk of these stocks meaning that if we say that there are three stocks in a Portfolio and their individual risk is 10%, 20% etc. then the risk of our Portfolio will not be calculated by adding the risks of all these three stocks and then divided by 3 but the risk of our Portfolio will further be less than the average of its individual risk and that is called impact of Diversification on our risk.

It means that risk of Portfolio is less than the average of risk of our individual stocks.

Why it is less?

As we have discussed in last lectures that it is because of the internal correlation of the stocks that exist among them and that also causes their covariance.

We can say that if there exist absolute positive correlation among the stocks of Portfolio then and only then the risk of the Portfolio will be the same as that of the average of the individual stock. It means that the average of value of the risk of individual stock will only be equal to the risk of the Portfolio if correlation among these stocks is +1 and practically there is no evidence that two stocks have got absolute positive correlation.

Portfolio's Risk

	Stock A	Stock B
Stock A	Covariance of A & A	Covariance of A & B
Stock B	Covariance of B & A	Covariance of B & B

If we find correlation of any one variable with itself then it will always be absolute 1 because in both columns there will be 100% identical values.

And if we find correlation of any one variable with any other variable then its value is not 1 but may be less than 1.

Portfolio's Risk

	Stock A	Stock B
Stock A	$\sigma_a \sigma_a \rho_{aa} X_a X_a$	$\sigma_a \sigma_b \rho_{ab} X_a X_b$
Stock B	$\sigma_a \sigma_b \rho_{ab} X_a X_b$	$\sigma_b \sigma_b \rho_{bb} X_b X_b$

σ_a is the standard deviation or the risk of individual stock A.

ρ_{aa} means that this is the value of correlation coefficient of stock A with stock A

X_a means the percentage of money or investment that we have put in stock A.

When we add these blocks then:

$$\sigma_a \sigma_a \rho_{aa} X_a X_a + \sigma_a \sigma_b \rho_{ab} X_a X_b + \sigma_a \sigma_b \rho_{ab} X_a X_b + \sigma_b \sigma_b \rho_{bb} X_b X_b$$

Equation

Risk of Portfolio

$$\sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b)$$

$$\text{Variance of Portfolio} = \sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b)$$

$$\text{Standard Deviation or Risk of Portfolio} = \sqrt{\sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b)}$$

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Topic 134: Portfolio Risk Analysis-3

We have discussed in previous lecture that risk of Portfolio is less than the average of risk of our individual stocks.

Why it is less?

It is because of the internal correlation of the stocks that exist among them and that also causes their covariance.

We can say that if there exist absolute positive correlation among the stocks of Portfolio then and only then the risk of the Portfolio will be the same as that of the average of the individual stock. It means that the average of value of the risk of individual stock will only be equal to the risk of the Portfolio if correlation among

these stocks is +1 and practically there is no evidence that two stocks have got absolute positive correlation.

We are continuing our discussion that if an investor increases diversification in his Portfolio meaning that if he further adds more shares or stocks in his Portfolio then what will be the effect on his risk due to it?

If a Portfolio is more diversified would it have a higher rate of risk or less?

Its answer is that more diversified is a Portfolio, lesser will be the risk of that Portfolio.

	Stock A	Stock B	Stock C
Stock A	<u>Cov. of A & A</u>	<u>Cov. of A & B</u>	<u>Cov. of A & C</u>
Stock B	<u>Cov. of B & A</u>	<u>Cov. of B & B</u>	<u>Cov. of B & C</u>
Stock C	<u>Cov. of C & A</u>	<u>Cov. of C & B</u>	<u>Cov. of C & C</u>

If we find correlation of any one variable with itself then it will always be absolute 1 because in both columns there will be 100% identical values.

And if we find correlation of any one variable with any other variable then its value is not 1 but may be less than 1.

	Stock A	Stock B	Stock C
Stock A	$\sigma_a \sigma_a \rho_{aa} X_a X_a$	$\sigma_a \sigma_b \rho_{ab} X_a X_b$	$\sigma_a \sigma_c \rho_{ac} X_a X_c$
Stock B	$\sigma_a \sigma_b \rho_{ab} X_a X_b$	$\sigma_b \sigma_b \rho_{bb} X_b X_b$	$\sigma_b \sigma_c \rho_{bc} X_b X_c$
Stock C	$\sigma_a \sigma_c \rho_{ac} X_a X_c$	$\sigma_b \sigma_c \rho_{bc} X_b X_c$	$\sigma_c \sigma_c \rho_{cc} X_c X_c$

σ_a is the standard deviation or the risk of individual stock A.

ρ_{aa} means that this is the value of correlation coefficient of stock A with stock A

X_a means the percentage of money or investment that we have put in stock A.

When we add these blocks then:

$$\sigma_a \sigma_a \rho_{aa} X_a X_a + \sigma_a \sigma_b \rho_{ab} X_a X_b + \sigma_a \sigma_c \rho_{ac} X_a X_c + \sigma_a \sigma_b \rho_{ab} X_a X_b + \sigma_b \sigma_b \rho_{bb} X_b X_b + \sigma_b \sigma_c \rho_{bc} X_b X_c + \sigma_a \sigma_c \rho_{ac} X_a X_c + \sigma_b \sigma_c \rho_{bc} X_b X_c + \sigma_c \sigma_c \rho_{cc} X_c X_c$$

Equation of Risk of Portfolio:

$$\sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + \sigma_c^2 X_c^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b) + 2 (\sigma_a \sigma_c \rho_{ac} X_a X_c) + 2 (\sigma_b \sigma_c \rho_{bc} X_b X_c)$$

$$\text{Variance of Portfolio} = \sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + \sigma_c^2 X_c^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b) + 2 (\sigma_a \sigma_c \rho_{ac} X_a X_c) + 2 (\sigma_b \sigma_c \rho_{bc} X_b X_c)$$

Standard Deviation or Risk of Portfolio =

$$\sqrt{\sigma_a^2 X_a^2 + \sigma_b^2 X_b^2 + \sigma_c^2 X_c^2 + 2 (\sigma_a \sigma_b \rho_{ab} X_a X_b) + 2 (\sigma_a \sigma_c \rho_{ac} X_a X_c) + 2 (\sigma_b \sigma_c \rho_{bc} X_b X_c)}$$

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Topic 135: Capital Asset Pricing Model CAPM-1

Capital Asset Pricing Model (CAPM) actually refers to the model which helps us to determine the return of a stock or a portfolio.

Capital Asset Pricing Model (CAPM) is also another method or direction to calculate the return of any stock or portfolio as we have discussed earlier.

Components of CAPM:

- R_f = Risk Free Rate of Return
- R_m = Market Rate of Return
- $R_m - R_f$ = Risk Premium
- β = Beta (Index of Systematic Risk)

R_f = Risk Free Rate of Return:

We have discussed earlier that Risk and Return are proportional to each other. It means that if risk increases then return will also increase.

If we eliminate risk then return will become zero or not?

No, practically there is a possibility of getting some return even with minimum risk. So, that means that a fixed return that an investor can get by not taking any risk is called **Risk Free Return**.

To practically define Risk Free Return, we can say that it is a return on T-Bills that is Treasury Bills, the return that we get on the bills issued by government which is the minimum actually generally taken as the **Risk Free Rate of Return**.

R_m = Market Rate of Return:

Market Rate of Return is second component of CAPM. How much return the investors are earning on average in the Stock Market? We can do this calculation

by Stock Market Index means that if we see that how much percent index of stock market goes up or down then we can also determine Market return through it.

A **well-diversified Portfolio** that is a Portfolio which is so large that it is almost equal to the market portfolio may also give us anticipated value of Market Rate of Return. This Market Rate of Return is always more than the Risk Free Rate of Return.

$R_m - R_f = \text{Risk Premium or additional return:}$

So, what is the difference Market Rate of Return and Risk Free Rate of Return and what do we call it?

When we take the difference of Market Rate of Return and Risk Free Rate of Return then the value that we get is called Risk Premium. It means that we have invested in the market without earning Risk Free Rate of Return and have taken Risk and the extra return that we will get on our investment due to this Risk which is called Risk Premium.

$\beta = \text{Beta (Index of Systematic Risk):}$

Beta of CAPM is actually the index of systematic risk meaning that Beta is also an integral part of CAPM. When we calculate the risk of an individual stock or portfolio through CAPM model then we also utilize the value of Beta.

So, we can conclude our discussion that CAPM model is actually the name of calculating the value of return with reference to the Risk Free Rate of Return and Market Rate of Return. But Beta (β) plays an important role in this model; if this beta is for market then its value will be 1 and if its value is more than 1 then it means that the return of our Portfolio or stock is more than the risk of market.

And if the value of beta will be less than 1 then the return of our Portfolio or stock is less than the risk of market.

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Topic 136: Capital Asset Pricing Model CAPM-2

Capital Asset Pricing Model (CAPM) actually refers to the model which helps us to determine the return of a stock or a portfolio.

Capital Asset Pricing Model (CAPM) is used to calculate the risk adjusted return on any stock or portfolio. We have seen in previous lectures that one calculation we have done through Average or Mean and then we said that we can also calculate the Expected Return through CAPM.

Components of CAPM:

- R_f = Risk Free Rate of Return (a fixed or minimum rate of return that an investor can get by not taking any risk)
- R_m = Market Rate of Return (that average return that can be earned in any security or in stock market, every investor can earn that average return of the market)
- $R_m - R_f$ = Risk Premium (the difference of Market Rate of Return and Risk Free Rate of Return then the value that we get)
- β = Beta (Index of measuring the Systematic Risk, we can see that the value of Beta tells us that the risk of any individual stock or any portfolio as comparison to the risk of market is more or less)

What is Security Market Line?

Security Market Line is such a graph or line which is made between two variables. On a graph paper, if we take return of market on the x-axis and return of an individual stock or a portfolio on the y-axis and plot a graph or different values in it then we get a graph or line from which we calculate the value of β . When we calculate the expected returns of different stocks by using the value of β which results in the form of SML Equation and the graph that we draw from it is called the Security Market Line or SML.

SML Equation:

The equation which is used in CAPM model we call this equation SML (Security Market Line) equation. And SML (Security Market Line) is a graph or line which is drawn in such a way that we take values of different β on x-axis and different values of return on y-axis. And the equation which we use to draw this line is called SML equation.

SML Equation:

$$\text{Return} = R_f + (R_m - R_f) \beta$$

Where

- R_f = Risk Free Rate of Return
- R_m = Market Rate of Return
- $R_m - R_f$ = Risk Premium
- β = Beta (Index of Systematic Risk)

If **β value** of any individual stock or portfolio **is equal to 1** then we will say that the risk of that stock is exactly equal to the risk of market. And if this beta is for market then its **value will be 1** and if its **value is more than 1** then it means that the risk of our Portfolio or stock is more than the risk of market. And if the **value of beta will be less than 1** then the risk of our Portfolio or stock is less than the risk of market.

If **value of β is less than 1** then our expected return will be less than the return of market and if the **value of β is more than 1** then our expected return will be more than the return of market. And if the **value of β is 1** then our expected return will be equal to the return of market.

We can conclude our discussion that SML equation actually is the equation of CAPM (Capital Asset Pricing Market) model and we calculate our Expected Return by using this equation and in this equation we have got R_m , R_f and value of β which will affect our Expected Return.

Module 7: Risk & Return

Topic 137: Fama and French Three Factor Model

We have discussed CAPM (Capital Asset Pricing Model) in which we have seen its components, Security Market line equation and also seen that how CAPM (Capital Asset Pricing Model) help us to calculate risk adjusted return.

But CAPM (Capital Asset Pricing Model) is an old Model and over the period of time other researchers have come up with better versions or better explanations of the expected return.

Fama and French Three Factor Model is the continuation of the CAPM (Capital Asset Pricing Model) where these two scientists added two more factors in the equation and they said that if these two factors are included then we get better explanation of our return or we can calculate our expected return in a better way.

As in CAPM (Capital Asset Pricing Model) we only have got one factor which affects our expected return and that is the Market Risk Premium ($R_m - R_f$ = Risk Premium) and we said that this is only factor that determines our expected return.

Fama and French have proved through their research that there are two other factors that affect our expected return that they name them Size and Value. So, in this factor model of Fama and French Three Factor Model, we have got three factors:

1. Market Risk
2. Size Risk
3. Value Risk

Market Risk:

Market Risk which we measure through β (Beta) and Market Risk Premium that we show through $R_m - R_f$

Size Value:

The second element that Fama and French added in their equation is *Size factor*. They have seen through their research that small size CAP companies in stock markets; the companies whose market capitalization is less where Market Capitalization is a term which is calculated as:

Market Capitalization = market price of a share x total number of outstanding shares

So, small capitalization or small CAP companies are those whose market capitalization is less and large or big capitalization or big CAP companies are those whose market capitalization is more.

Fama and French have seen that small CAP companies perform better than large CAP companies and they created a variable which they named:

Size = Small Cap – Big Cap

The size of a company effect on the return so, they consider it as a variable in addition to the Market Risk Premium.

Value Risk:

The third factor that they considered was the Value factor. Here Value means that not market value or book value but a ratio of book value to market value. If ratio of the book value to market value of any stock will be high then we will say that it is more valuable stock.

Book to market value means that if we divide book value with the market value or price of that stock and as a result we get a value which may be equal to 1 or more than 1 or less than 1.

Fama and French have seen that whose book to market value is high and its return will also be high and this is the factor that affects the return. So, they determine high value stocks whose book to market value is high and whose book to market value is low that they call low value stocks. And they created a new variable which is Value variable which is calculated as:

Value = Return of **High value stock** – Return of **Low Book to Market Value stock**

So, **Fama and French** came with this equation which is the **three factor model equation of Fama and French**. If we this equation, this is a Regression equation (the equation in which independent variables that are written on right side of equation effect the value of variables written on the left side of equation and we call such kind of equation, a regression equation) as we said that **CAPM equation is straight line equation** and we have more similar equation given below:

F & F Three Factors Equation

$$\text{Return} = R_f + (R_m - R_f)\beta_1 + \text{Size}\beta_2 + \text{Value}\beta_3$$

OR

$$\text{Return} = R_f + RP\beta_1 + \text{Size}\beta_2 + \text{Value}\beta_3$$

Here **Return** is dependent variable which depends on the market risk premium, size and value according to the Fama and French Three Factor Model.

If we change the value of Size and Value and Risk Premium then our Expected Return will also be altered or changed.

In CAPM, we have one Beta value because we have just one independent variable that was **Market Risk** but in **Fama and French Three Factor Model** we have *three independent variables or three factors* so, we need to have three Beta values and we simply differentiate them by calling them β_1 , β_2 and β_3 where β_1 will adjust our risk premium, β_2 will adjust the variable of factor of size and β_3 will adjust the variable of value.

So, we can conclude our discussion of **Fama and French 3 Factor Model** that Fama and French presented a better equation or model to calculate Expected Return or Risk adjusted return as compare to CAPM in which they introduce two more factors and said that not only Market Risk Premium affects the occurring changes in our return but Size and Value also affects it.

Module 7: Risk & Return

Topic 138: Four Factor Model

This lecture is about the Four Factor Model. This Four Factor Model is also called the Carhart Four Factor Model

To understand this model, we need to link this model with the previous Three Factor Model that we have learnt in the last lecture and that is called the Fama and French Three Factor Model. In this three factor model, we have talked about two new variables in addition to the Market Risk and that is the Size factor and Value factor.

In this Four Factor Model which is the Carhart Four Factor Model, we will talk about another variable that is called the 'Momentum' variable.

Momentum is actually about the prices of stocks going up or the prices going down which actually creates the Return on a particular stock and also the Risk on that stock. We have discussed it earlier that the Risk is proportional to the Return. So, whenever we use these terms Risk and Return, these are directly proportional so, interchangeably we use these terms.

Momentum:

Here we consider the stocks or the shares whose prices are going up and we call them Winners. So, Winners are simply the stocks which are gaining in the stock market. And against stocks' Winners there are losers; the stocks whose prices are going down. So, we construct a variable by the name of Momentum by taking the difference between Winners and Losers. We also refer them to Ups and Downs so the Momentum variable is given below:

$$\text{Momentum} = \text{Winners} - \text{Losers (Ups - Downs)}$$

And Carhart actually gave this theory, this model which was then implemented and it was seen that by including this Fourth Factor of Momentum, there is a better explanation of the return on the stocks.

Momentum Defined:

“The tendency for the stock price to continue rising if it is going up and to continue declining if it is going down”

In the stock market, it is considered which is based on the findings and results of the price movement then when a stock price starts moving up it gets a momentum and as a result it keeps on going up to a certain level where that momentum is then broken and the stock price starts falling down.

And whenever the stock price starts falling down, it doesn't stop immediately and as a result what happens that a momentum is again there because of this price is continuous to fall down. This momentum of prices going up or down meaning the stocks getting going up or coming down in terms of their prices which are termed as **Winners or Losers** create this variable of Momentum.

The Equation of Carhart's Four Factor Model

$$\text{Return} = R_f + (R_m - R_f)\beta_1 + \text{SMB } \beta_2 + \text{HML } \beta_3 + \text{UMD } \beta_4$$

We have talked about the first three factors in detail, the first factor which is that of the **CAPM (Capital Asset Pricing Model)** is Market Risk Premium which is the difference of Market Rate of Return and Risk Free Rate of Return ($R_m - R_f$) and then multiplied by the Beta value which in this model is the first beta or β_1 .

In this three factor model, we have got a Size factor (where, Size = Small – Big) multiplied by β_2 and Value factor (where, Value = High book to market value stocks – Low book to market value stocks) multiplied by β_3 and now we added fourth factor or variable which is the addition or contribution of Carhart in this model and we have said that addition of this variable has actually improved the return's definition meaning that adding Momentum (**Momentum = Ups minus Downs or Winners – Losers**) actually provides a better explanation of the return on investment.

So, concluding our discussion on Carhart Four Factor Model, we can say that when Fama and French introduced two additional variables or factors that is Size and

Value, we got a better explanation of Return *than the CAPM*, those two variables combined with the Market Risk Factor explained the Return of an investor in a better way.

And when **Carhart fourth factor** that is *Momentum was added*, it actually provided even a better explanation. So, we can say that the Carhart Four Factor Model is a model that gives or explains the Return of the investor better than Fama and French Three Factor Model.

Module 7: Risk & Return

Topic 139: Multiple Factors Risk Adjusted Models

As you remember that when we have start our discussion on **CAPM (Capital Asset Pricing Model)**, we talked about the calculation of Return considering the Risk Free Rate of Return and the Additional Risk or Risk Premium taken by an investor by putting his money in the stock market and that created a Risk Premium.

And then we talked about adjusting that Risk Premium considering the Beta (β) value where β is the index of systematic risk meaning that the index of the risk or the instrument to measure the risk present in the market. And we will also discuss further about the value of β in upcoming lectures.

We have continued our discussion on CAPM and we look forward to the **Three Factor Model of Fama and French** where we introduce the Size and Value factors and ultimately we added the fourth factor and that fourth factor was **Carhart Four Factor Model** where Momentum was also considered.

And we saw that this is what the research actually proved that adding the fourth factor of Momentum provided a better explanation of Risk adjusted Return so, we have said that **Carhart Model** has further improved Fama and French Three Factor Model.

But the researchers have not stopped at Four Factor Model and explored many other factors. And it has been seen that can we further improve the model by using

these factors? Is there a better explanation of the risk adjusted return including other factors?

And as a result we came up with several factors model that is Multiple Factor Models and there are many different Multiple Factor Models which were developed by different researchers and financial experts.

We have to see different variables in these Multiple Factor Models and do experimentation with them and see that is our explanation of the model has increased because we know that we have talked about an equation of CAPM and also that was then taken forward in the Three and Four Factor Model, is a regression equation where the independent variables that is Market Risk, Size, Value and Momentum they all explain the dependent variable which is Return.

If the variable which we include in our model will explain our dependent variable more and more that in statistical term we show it by R^2 meaning that this R^2 tells us that in our model, the included independent variables (Market Risk, Size, Value and Momentum) in combination how much do they explain the dependent variables.

It means that if the value of R^2 comes 70% or 0.7 which means our model is 70% explaining the variation of the dependent variable. And we would like certainly to have more and more explanation of the dependent variables.

Now, the question is that when we further add more factors in our Four Factor Model then are dependent variables the further explanation of return that is further the value of R^2 increases or not?

If the value of R^2 decreases instead of increase by including any factor then we will said that this factor is not relevant for us and there has no improvement in the model by its usage.

But researchers have worked on different factors and they have come up with different results. Therefore like CAPM and Three Factor Model, we cannot say anything conclusively that which of the Multiple Factor Model provides the best explanation.

And this is not the only thing; there are different types of funds or investments whose explanations are provided by different types of dependent variables.

Multiple Factors or Variables:

Lets have a look that what are these Multiple Factors or Variables which are actually included in these models.

- Macroeconomic
- Fundamental
- Statistical
- Combination

Macroeconomic: Some of them are Macroeconomic variables. Macroeconomic simply means that the economics which is related to the country as a whole or economy as a whole. For example, the GDP (Gross Domestic Product), the trade balance, the balance of payment and all macroeconomic variables.

Fundamental: It means the keen indicators of our portfolio or companies which show the strength and weaknesses of that company. For example profitability, liquidity and similar other variables.

Statistical: There are some such variables which are statistical and these are those which are based on the calculation that is done to get some values and develop these variables.

Combination: Finally there are variables which are combination of all these factors or variables.

Fung Hsieh 7-Factor Model

To see the Return and to see the explanation of Return on Hedge Funds, Fund Hsieh 7-Factor Model is also a popular model. If you look at these factors then you would find that none of these 7 factors is out of those which we have discussed in the Three Factor Model or in the Four Factor Model as given below:

Hedge Fund Returns

1. Bond Trend Following
2. Currency Trend Following
3. Commodity Trend
4. Equity Market
5. Equity Size
6. Bond Market
7. Bond Spread Size

The reason is that this model is specifically designed to see the return of Hedge Funds.

The Equation of Multiple Factor Model

$$\text{Return} = R_f + (R_m - R_f)\beta_1 + \text{SMB } \beta_2 + \text{HML } \beta_3 + \text{UMD } \beta_4 + \dots + N \beta_n$$

Now, we can conclude our discussion that Multiple Factor Model will be effective or useful if these factors in addition to the four factors which we have talked about can better explain the Return of an investor on a particular fund or on a particular stock or on a particular portfolio.

Module 7: Risk & Return

Topic 140: Calculation of Beta using Characteristic Line Index of Systematic Risk

In this lecture, we will learn about calculation of β (Beta) using a Characteristic Line that is graphically and also using a statistical Formula.

What is β (Beta)?

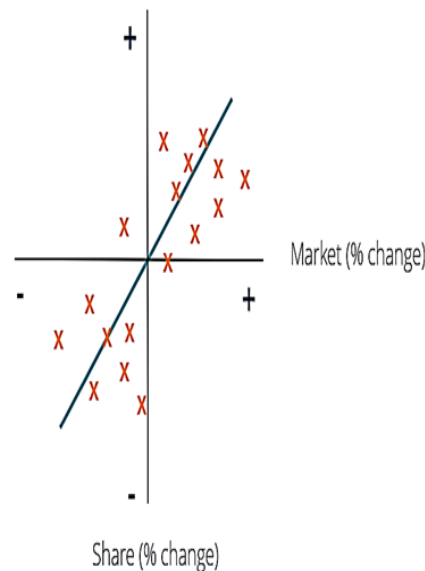
We have discussed about β (Beta) while discussing CAPM (Capital Asset Pricing Model) and we have said that β (Beta) is defined as the Index of Systematic Risk and Systematic Risk itself is defined as the Risk which is there in the market meaning that the risk of market is called the Systematic Risk.

And we explained it as this is basically the Risk which is Unavoidable. The risk which is something that cannot be eliminated or reduced by considering the process of diversification.

The tool or instrument that we use to measure the built-in risk in the market, is called β (Beta). β (Beta) is actually the Slope of Tangent where Slope of Tangent means that there is a line or graph which is made between two variables and the slope of that graph or inclination of that graph is called β (Beta).

Characteristic Line

Beta = slope of the line



In an equation of Regression which is also called the Equation of Straight Line, we have got two variables; one variable which we consider on x-axis and other one variable which we take on y-axis. The variable taken on x-axis is always Independent Variable and the variable taken on y-axis is always Dependent Variable.

Independent variable is actually such variable whose change also makes changes in the other variable that is why we say that a **Dependent variable** is dependent on the changes in the Independent variable.

Now, considering this, if we look at this graph we see that on x-axis we take the Market Return meaning how much return is there in the market at any point of time i.e., if we are calculating daily data, on a given day at the end of the trading in the stock market, we will see that how much was the return on the market portfolio and in against that what was the return on the stocks or portfolios' return which was on the y-axis.

And we place a point on the graph and similarly, we continue to draw points between the daily market's return or portfolio or stocks' return that we take on y-axis. And when we have a lot of that data which is called a **Scattered Diagram** and we pass a straight line between it and that straight line is passed in such a way that about 50% of the values are above that line and about 50% of the values are below that is the line passes through the center of all these values.

Now, then we look at the percentage change in x-axis and the percentage change in the y-axis over that line. It means that when we move from point 1 to 2 on x-axis considering that graph or line which is a characteristic line then from the point 1 to on which point and what change has happened on the y-axis.

If the change in y-axis is also from point 1 to 2 then we will say that the change of x-axis and y-axis is equal. This is shown as ∂x or dx and the change on the

Characteristic Line is shown as the first Differential that we show as $\frac{\partial y}{\partial x}$ or $\frac{dy}{dx}$ meaning that how much change has happened in y due to the change of x.

And this is how we calculate the value of β (Beta).

So, β (Beta) is actually $\frac{\partial y}{\partial x}$ or $\frac{dy}{dx}$ meaning that the change occurring in y with reference to the change in the x.

The **Statistical Formula** of β (Beta) is given below:

The Formula



Beta Formula =
$$\frac{\text{Covariance (Re, Rm)}}{\text{Variance (Rm)}}$$




R_m = Market Return

R_e = Return on the portfolio or the stock

So, we can calculate β (Beta) graphically by putting down the points at the graph paper by considering the return of market and that of the portfolio or of the individual stock and then passing a Characteristic Line from the middle of those points and then taking two different points on those lines and dividing that value of y with x and getting the value of β (Beta).

But we also have a statistical formula to calculate β (Beta) where we have the Covariance of the Market return and the return on portfolio or stock divided by the Variance of the Market Return.

Covariance is actually the Variation in two values with reference to each other and we have seen that this Covariance depends on the **Correlation** of these two variables.

So, we conclude our discussion, we say that β (Beta) which is the index of systematic risk is something which can be calculated graphically by drawing a Characteristic Line and then taking the relationship of change in y and change in x or the value of β (Beta) and also be calculated by using the Statistical Formula that is the Covariance between the two stocks or two variables that is Market and portfolio's return and dividing it by the β (Beta) Variance of Market Return.

Module 7: Risk & Return

Topic 141: Calculation of Portfolio Beta

We have discussed that **β (Beta)** is the index of systematic risk and it is the relationship of Market Return with the Return of an individual stock or the return of a portfolio. It means that we can take any individual stock's return or also can take any portfolio's return on the y-axis.

The **Portfolio** is nothing but a group of various stocks meaning that if we combine several stocks and make investment in them which will be called our Investment Portfolio.

A **Mutual Fund** is a very large Portfolio in which many stocks are included. If we compare that portfolio's return with the market's return and calculate the value of β (Beta) then this will be called the β (Beta) of that portfolio.

And if we have the β (Beta) of an individual stock and calculate the value of β (Beta) for an individual stock then we will consider the returns of that individual stock on the y-axis and that will be the β (Beta) of that individual stock.

Stock Beta Values

Now, Individual stock can have Risk or Return equal to Market, more than the Market or less than the Market.

Beta Less than 1

If the β (Beta) of a particular stock is less than 1, it means that the change in the y variable is less than the change in x variable and that indicates that β (Beta) is less than 1 which indirectly means that the risk of the stock is less than the average risk of the market. It means that if the value of β (Beta) is less than 1 then it simply means that return of that stock will be less than the return of the market.

Beta Equal to 1

It will be very rare that some stocks have a β (Beta) exactly equal to the market β (Beta) and its value of β (Beta) becomes 1. Here the value 1 means that the change occurring on the x-axis is exactly equal to the change occurring in the y-axis. It means that there is a return in the market and exactly equal return will be on that

stock. So, we say that β (Beta) of that stock will be 1 which means that the risk and return of that stock will exactly equal to 1.

If we relate market return with the market return meaning that we have market return on x-axis and also have market return on the y-axis then we can understand that values of that variable will be same on both x-axis and y-axis. And in that case, we will get exactly 1 value of β (Beta) that is the relative change in x will be exactly equal to the relative change on the y because we have taken same variable on both axes. So, any such stock which has taken on y-axis and its return is exactly equal to the market's return then we can understand that it is difficult or may be rare.

Beta More than 1

There is also a possibility of having a β (Beta) value more than 1. Here the value of β (Beta) more than 1 means that the change on y-axis that is ∂y or dy is more than the change on x-axis that is ∂x or dx meaning that when we see the change of return of stock then is more than the change of return of the market. If there is 1% increase of return in the market then return of that stock will be more than 1%.

**Portfolio
Beta**



We calculate the individual stock's β (Beta) value. If we have any portfolio's combined return then we can very easily that we can determine the β (Beta) value of that portfolio by regressing that combined return with the risk of market or by taking the portfolio's return on y-axis or by taking the market' return on x-axis.

But if we have individual stock's β (Beta) but do not have combined β (Beta) of portfolio then can we calculate the β (Beta) value of portfolio?

The answer is Yes. And that portfolio's β (Beta) will be nothing but the weighted average of all the β (Beta) values in the portfolio. **Weighted average** is nothing but we average the values of β (Beta) and consider the percentage of investment in that stock.

Company	Beta	Percentage of Investment
A	1.2	20%
B	1.3	30%
C	1.1	40%
D	0.9	10%

Calculations

$$(1.2 \times 20\%) + (1.3 \times 30\%) + (1.1 \times 40\%) + (0.9 \times 10\%)$$

$$= (1.2 \times 0.2) + (1.3 \times 0.3) + (1.1 \times 0.4) + (0.9 \times 0.1)$$

$$\beta = 1.16$$

here the value of β (Beta) is more than 1 so we will say that it is possibility that the expected return on our portfolio is 16% more than the return of market meaning that if the market will earn 10% return, our portfolio is likely to give us 11.6% return.

Module 7: Risk & Return

Topic 142: Security Market Line

SML (Security Market Line) is actually a line which is drawn between two variables. As we have discussed SML when we were discussing about CAPM (Capital Asset Pricing Model) where we have learnt that the equation of CAPM is actually the SML (Security Market Line) equation.

To understand SML, we need to remember or recall that CAPM equation. In CAPM equation, we take return of return on one side which is the expected rate of return and that is the **dependent variable**. And on the right side of the equation, we take the Risk Free Rate of Return which is actually the intercept.

Here **intercept** is actually that value on y-axis while on x-axis value becomes zero meaning that when we take one variable on x-axis and takes another variable on y-axis and when on x-axis the value of variable is zero then if the value of variable on y-axis becomes zero then Intercept becomes zero.

But when the zero value of variable taken on x-axis and any value of variable taken on y-axis exists then that value called **Intercept**. If independent variable x becomes zero then there will exist some value of dependent variable and that is the equation of CAPM which is as:

SML Equation:

$$\text{Return} = R_f + (R_m - R_f) \beta$$

Where

- R_f = Risk Free Rate of Return
- R_m = Market Rate of Return
- $R_m - R_f$ = Risk Premium
- β = Beta (Index of Systematic Risk)

Suppose, in that equation if Risk Premium is zero then the whole factor $(R_m - R_f)\beta$ will be equal to zero and in that case the expected return is equal to Risk Free Rate of Return (R_f) as:

$$\text{Return} = R_f + (R_m - R_f) \beta$$

$$\text{Return} = R_f + (0) \beta$$

$$\text{Return} = R_f$$

It means that any investor's return is not zero even if his risk level becomes zero and that is why R_f is the Risk Free Rate of Return. And this R_f is actually the intercept. That R_f indicates that value which will be the return of investor without taking a Risk.

So, how do we make this equation? How do we actually regress this Independent Market Rate of Return or Risk Premium with the dependent variable of the return that is through what is called Security Market Line (SML) equation.

Security Market Line (SML) is actually a graph or line which is drawn two variables and these two variables are β (Beta) which is an independent variable and the return of the investor. It means that while drawing SML, we consider different values of β (Beta) on x-axis meaning that if the value of β (Beta) will be 1 then what will be the return of the investor and if the value of β (Beta) will become 1.1 then what will be the return of the investor and if the value of β (Beta) will become zero then what will be the return of the investor.

As now we have discussed that even if the β (Beta) values becomes zero that does not mean the return of the investor will be zero. But it will be minimum and that minimum rate of return is called the **Risk Free Rate of Return (R_f).**

Betas	Returns
0.7	6%
0.8	7.50%
0.9	7.90%
1	8.40%
1.1	8.90%
1.2	9.70%
1.3	10.30%
1.4	11.20%
1.5	12.80%
1.6	14.50%

$$\text{Return} = R_f + (R_m - R_f) \beta$$

We have calculated the values of Returns in above table by putting the value of Betas in the equation of SML (Security Market Line) where R_f and $(R_m - R_f)$ will remain constant and as the Beta will get changed then value of equation will also get changing.

Let's have a look at this graph which is actually the SML (Security Market Line).



So, we conclude our discussion, **Security Market Line (SML)** is a line which is between different values of β (Beta) and the expected return. We can get these values by using the equation of Security Market line (SML) which is also called the CAPM equation in which we put different values of β (Beta) in the constant value of Risk Free Rate of Return (R_f) and constant value of Risk Premium ($R_m - R_f$) and get values of expected returns.

We present these values on the graph and the line that we get that is called **Security Market Line (SML)** and it simply shows the expected return of an investor with different values of β (Beta).

Module 7: Risk & Return

Topic 143: Arbitrage Pricing Theory

So far we have talked about various **Risk Adjusted Return Models** as we have talked about **CAPM (Capital Asset Pricing Model)** which has been developed many years ago in which it is told that the **Risk Premium of market is only a factor that should be used to calculate the expected return.**

The only factor that explains the return of an investor and it was a successful discovery of researchers and **CAPM became the model which was very commonly used by the investors.** Then different researchers have to add more factors in this model and test it every time.

We have seen **Fama and French Three Factor Model, the Carhart Four Factor Model** where we added another factor which is called **Momentum** and we have also said that **if we further add different factors or variables in the model then is our expected returns values becomes more better or not.** Is our expected return be more better explained by adding these factors or variables?

In previous lectures we have talked about that the **meaning of the explanation of expected return is that our expected return depends on which things.** If we use any regression equation or model and value of R^2 becomes 100% which is not possible in any statistical model and it can only be possible in any mathematical model that 100% factors has to be included which are showing the variation of dependent

variables meaning that our expected return change due to which things. If we observe all these things and include them in the model then our R^2 which tells us that how much percent the change of our expected return is dependent on the independent variables and its value will become 100% meaning that we have accounted all those factors.

But in behavioral finance where we understand various factors in the variations of stock market or investors are not possible to find out all the factors and include them in a model to get 100% explanation of the variable. So, we include different variables in our model and see that is our expected return getting better or its explanatory power of that model is increasing or not. Here the increase of explanatory power means that our R^2 is increasing or not. So, we keep on adding different variables or factors.

In Multi Factor Risk Adjusted Models, we said that people have developed different kinds of models for different types of portfolios as we learnt about Fung's Model which is used to explain the return of Hedge Fund and independent variables in it were very different from the CAPM and other 3 or 4-Factors models.

We take forward this discussion that we have a theory that we name Arbitrage Pricing Theory (APT).

Arbitrage Pricing Theory (APT) is actually the continuation or form of Multi-factor Risk Adjusted Returns Models that we use.

Process of Arbitrage:

Arbitrage means that if there are two items/two stocks/two shares available in the market which are actually providing us the same rate of return but the price of one is more than the other then the process of Arbitrage takes place. It means that people will go towards the cheap shares by selling expensive shares as a result sale of expensive share will get started and due to that selling pressure, its price will get falling down.

So, Arbitrage is actually a process in which if it is assumed that there exist two such stocks/commodities/shares in the market whose return is same but price or value is different then every investor will want to purchase cheap share or less

valued share in order to get same return. And as a result, a buying pressure will be created for the stock which is of low value and a selling pressure will be created for a stock which is got a higher price. So, the process of Arbitrage is selling the expensive share or commodity and buying the less expensive share or commodity keeping our return the same.

Now, Arbitrage can be done in different ways; it means that when we get change one thing with other thing and see that if our return or understanding or explanation is getting increasing from it then we will take in the term of Arbitrage.

For example, if we exclude any one factor or variable from any Multi-factor model then its explanatory power decreases or increases. And that is basically the process of Arbitrage.

So, this Arbitrage Pricing Theory (APT) actually states that if we keep on adding and deleting various factors in the Multi-factor model and we try to see that where the model's explanation of the dependent variables or the return is increasing and decreasing, we can come up with a model or variables which will best explain the return on that stock or portfolio.

So, we can conclude our discussion that Arbitrage Pricing Theory (APT) is such a theory which tells us that we add such factors or variables through replacement in Multi-factor model where our expected return is getting best explained and that is the most suitable model for that type of stock or that portfolio.

Module 7: Risk & Return

Topic 144: Efficient Market Hypothesis

Efficient Markets are those Markets which are considered to be efficient in terms of processing the information. Here the efficiency means that how much early we complete our work and this is the general definition of efficiency. When a person does his/her work in lesser time then we say that this person is more efficient.

Now, the stock market or security market can also be more or less efficient. This efficiency of stock markets or security markets can be in term of transaction meaning that trade (buying and selling) in these markets so these transactions or trades that is buying or selling of the stocks in the stock or security markets could be efficient. Such markets in which trading is efficient then we will call this market transactionally efficient market.

But in this hypothesis which is the Efficient Market Hypothesis, we are not talking about the transactionally efficient markets but we talk about **Informationally Efficient Markets** meaning that those stock or security markets which process information more quickly and even too much quickly that any investor cannot earn extra-ordinary return due to information. It means that when we say that any market is Informationally efficient then is the presence of information for any investor can provide more market's average return or any other investor's return?

How it could be possible?

Suppose, if I know that one company is going to give dividend then there is a chance that its share's price goes up as its giving dividend and if we have that information and purchase that company's share and as that dividend will be declared then this share's price will go up and we can earn more profit from that share.

But if this market will be informationally efficient then the information that we have that dividend is going to declare then this information will be spread in the market very quickly within seconds.

And the influence of this information (the increase of the price of this stock/share) on the price of that stock in the market will be too much quickly implemented that any investor cannot get extra-ordinary benefit from purchasing this stock by using this information. So, **efficient markets** are those markets which process all that information very quickly.

Types of Information:

We define information in two categories; one which is **Publically available information** meaning that such information which any investor or person can

access very easily (i.e., information published in the newspaper, information available in the Print Media or Electronic Media, information available on the website) and there is no restriction and such information is called the Publicly Available Information.

Other type of Information is **Inside Information**; such information which is not available to everyone or everyone cannot access that information but only a small number of people know this. For example, the people sitting in the Board of Directors of the company probably know something which is not yet been declared officially by the company. So, the information that the Board of Directors have is called the **Inside Information.**

When we say that Markets are efficient in terms of processing the Publicly Available Information and the Inside Information so these are considered to be the most efficient markets.

Efficient Market Hypothesis is a hypothesis. When we do any research and assume something or make hypothesis and then empirically prove such hypothesis true or false through data collection or analysis or experiments.

So, **Efficient Market Hypothesis** is also a hypothesis which states that markets are **efficient** in processing information. And any information that any investor he has, cannot earn an extra-ordinary profit through this information and that is called the Efficient Market Hypothesis.

There are **three Levels or Forms Efficient Market Hypothesis** on which we see that this hypothesis is application:

Weak Form:

The **Weak Form of Efficient Market Hypothesis** states that the past data, the data which is there in the trends in the past analysis meaning that all we have today and have been happened in the past or those prices which have been changed from today or those trends which we have made in the past which the technical analysts use in their analysis and decision making.

Weak Form of Efficient Market Hypothesis says that all that past information is reflected in the current market price. So, anybody who is analyzing the trends, the

past data or past information cannot make extra-ordinary profits in the stock. So, this is the **Weak Form.**

Semi Strong Form:

The second form or level is **Semi Strong Form of Efficient Market Hypothesis** in which there is not only negation of the information in the past data or past trends but that information which is publically available or which can be accessed by anybody or any investor is also negated. It means that anybody who is analyzing the past trends or has collected the publically available information cannot earn extra-ordinary profits.

So, somebody who says that I am a fundamental analyst and I have got a lot of information about the stock market and I can get extra returns, the **Semi Strong Form actually negates that.**

Strong Form:

This level or form of **Efficient Market Hypothesis** is Strong which negates the third form of information that we call Inside Information. It states that anybody who has got a past data, who has got the publically available data and even if the person has got Inside Information still would not be able to make extra-ordinary profits. Because all these information are reflected in the current market price.

We can conclude our discussion that researchers have proved this thing to the extent that **Weak Form of Efficient Market Hypothesis is correct to more extent that is past trends are reflected in the current market price.**

And there is also sufficient evidence that **Semi Strong Form of Efficient Market Hypothesis is also true** meaning that publically available information which can be related to company's growth, contracts, agreement or printed in the annual reports also cannot generally give us extra-ordinary information.

But there is very little evidence in the favor of Strong Form or Level of Efficient Market Hypothesis that Inside information cannot give us extra-ordinary information.

Usually it has been seen that the people having Inside Information can earn more return as compare to the normal investors.